Bao Nguyen	ECE332 Homework 3	05/12/26

1.) A 200 MHz left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air on a dielectric medium with $\epsilon_r = 4$ and occupies the region defined by $z \geq 0$. (20 points total)

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$. (5 points)

(b) Calculate the reflection and transmission coefficients. (5 points)

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \leq 0$. (5 points)

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

2. Repeat the above problem but replace the dielectric medium with a poor conductor characterized by $\epsilon_r = 2.25$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m (20 points total)

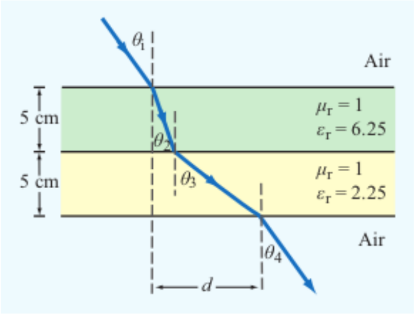
(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$. (5 points)

(b) Calculate the reflection and transmission coefficients. (5 points)

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \leq 0$. (5 points)

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

3. A parallel-polarized plane wave is incident from air at an angle θ_i = 30◦ onto a pair of dielectric layers, as shown above. (10 points total)



(a) Determine the angles of transmission θ_2, θ_3, and θ_4. (5 points)

(b) Determine the lateral distance d. (5 points)

4. A TE wave propagating in a dielectricfilled waveguide of unknown permittivity has dimensions a = 5 cm and b = 3 cm. If the x componentof its electric field is given by

$$E_x = -36\cos(40\pi x)\sin(100\pi y)\sin\left(\left(2.4\pi\times 10^{10}\right)t - 52.9\pi z\right)\ \ (\mathrm{V/m}),$$

determine: (20 points total)

(a) the mode number, (5 points)

(b) ϵ_r of the material in the guide, (5 points)

(c) the cutoff frequency, (5 points)

(d) and the expression for H_y. (5 points)

5. A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with a = 3 cm and b = 2 cm. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end? (8 points)